
Trans-cGAN: A Combined Framework for Text and Image Synthesis

Junyi Xu
Jacobs School of Engineering
University of California, San Diego
jux022@ucsd.edu

1 Introduction

The assignment requires designing a deep generative model capable of producing a full-length story and a representative image from a short textual prompt. Based on the knowledge I have learned, I propose a combination model of Transformer and Conditional Generative Adversarial Network (cGAN). Instead of a single "black box" model, I decouple the task into:

- (1) a **Text Module** using a Transformer to expand the prompt.
- (2) a **Image Module** using a cGAN to synthesize images grounded in the generated narrative.

2 Methodology

The information flow through our system is sequential:

Prompt $\rightarrow$ Transformer $\rightarrow$ Story $\rightarrow$ Encoder $\rightarrow$ cGAN $\rightarrow$ Image

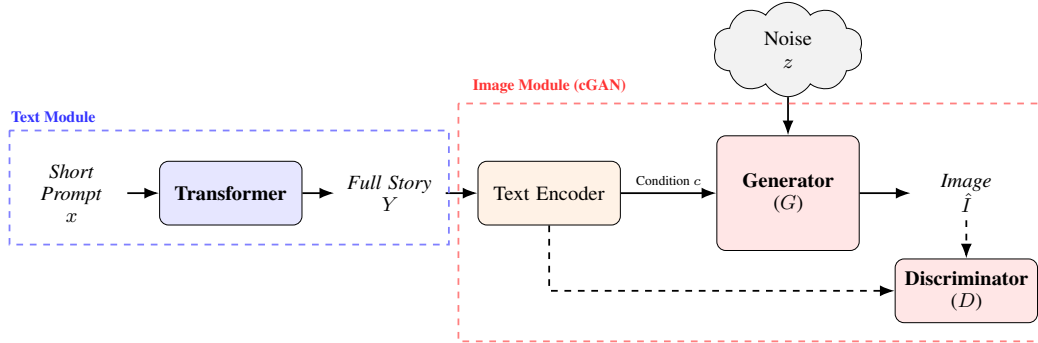


Figure 1: Architecture of Trans-cGAN. The model expands a short prompt into a full story using a Transformer, encodes the narrative into a condition vector c , and employs a cGAN to synthesize a coherent image from noise z (enhance the generalization ability of G) and the condition c .

2.1 Component 1: Text Expansion (Transformer)

Since the task only requires text extension, I only employ a Decoder-only Transformer for the text generation stage. Given a short prompt x , the model generates a sequence of tokens $Y = (y_1, \dots, y_T)$ autoregressively.

The core mechanism enabling this is Self-Attention, which allows the model to attend to relevant parts of the generated context [1]:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

The objective is to minimize the loss function: the negative log-likelihood of the story sequence:

$$\mathcal{L}_{text} = - \sum_{t=1}^T \log P(y_t | y_{<t}, x) \quad (2)$$

2.2 Component 2: Image Synthesis (cGAN)

Once the story Y is generated, I use a pre-trained text encoder (Bidirectional Encoder Representations from Transformers, BERT [2]) to compress it into a semantic feature vector c . This vector acts as the condition for the GAN. The cGAN framework is shown below:

- **Generator (G):** Maps a noise vector z and condition c to an image space: $\hat{I} = G(z, c)$.
- **Discriminator (D):** Evaluates whether an image is both realistic and matches the text condition c .

The training objective is the adversarial minimax [3]:

$$\min_G \max_D V(D, G) = \mathbb{E}_I [\log D(I|c)] + \mathbb{E}_z [\log(1 - D(G(z|c)|c))] \quad (3)$$

3 Training Strategy

Training such a hybrid system end-to-end is unstable. I propose a decoupled training strategy:

1. **Stage 1 (Text):** Train the Transformer on story datasets (e.g., BookCorpus [4]) using Teacher Forcing. The goal is to learn narrative structure and vocabulary.
2. **Stage 2 (Image):** Freeze the Transformer. Train the cGAN using pairs of (Image, Caption). The Generator learns to map text embeddings to visual features, while the Discriminator enforces visual realism.
3. **Inference:** During testing, the two modules are chained. The output of Stage 1 becomes the input condition for Stage 2.

4 Conclusion

The core of my idea is to divide the task into two parts: the first one is the text extension using a Transformer and the second part is to generate images that match the context of the extended text by utilizing a cGAN. My idea might have some errors or practical issues, which I might find out after conducting experiments.

References

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